AIRBUS	SYS	Ref: S2NG-SP-ADSF-SYS-1002224374 Issue: 01 Date: Mar 04, 2026	S2NG

Thermal Hardware Specification

This document does not contain national, EU or/and US export controlled technology (data)

Approval Information:

Name/Role:	Date:
SPEIDEL, Christopher (Airbus DS) [DE] /	Mar 05, 2026 03:06:38 PM GMT
GLATT, Marcus (Airbus DS) [DE] /	Mar 05, 2026 01:34:15 PM GMT
THEUNISSEN, Martijn (Airbus DS) [DE] /	Mar 05, 2026 12:37:19 PM GMT

AIRBUS certifies that the above persons have signed this document by electronic validation process supported by AIRBUS tools.
The current issue is the electronic copy available through Airbus Defence and Space PDM. All paper copies are for information only.

Left intentionally blank

Sentinel-2 NG

Thermal Hardware Specification

DRL-No.: Enter DRL-No.

CI-No.: Enter CI No.

Prepared by			Date:	
	Name / Function:	Christopher Speidel / Thermal Architect		
Checked by			Date:	
	Name / Function:	Martijn Theunissen / Chief Engineer		
Released by			Date:	
	Name / Function:	Marcus Glatt / Project Manager		

Approval and release information:

This document has been approved either by manual signatures as shown above, or via electronic workflow as shown on the cover page preceding this page.

Export Control Regulation:

See Page 2 for Export Control Information concerning this document's content.

Document autogenerated from DOORS

Module: Thermal Hardware Specification

Baseline: 1.0

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

EXPORT CONTROL INFORMATION

Section 1 (not applicable in France, please go to section 3)

This document contains Technical Information**:**

Yes ☑ **No** ☐

If No to section 1: Please complete section 2

If Yes to section 1: Please complete section 3 as applicable

Section 2 (not applicable in France, please go to section 3)

I confirm the document does not contain Technical Information and is **"Not-Technical"**.

Name:

Date:

Section 3

3a. National and EU regulations Export Control Assessment

This document has been assessed against applicable export control regulations in

☐ France ☑ Germany ☐ Spain ☐ UK ☐ Other:

☑ and does not contain Controlled Technology[1] and is therefore **"Not Listed / Not Controlled"**.

☐ and contains Controlled Technology with export control classification 9E001.

Note: Any transfer of this document in part or in whole must be made in accordance with the appropriate export control regulations. Prior to any transfer outside of the responsible legal entity, confirmation of an applicable export licence or authorisation must be obtained from the local Export Control Officer (ECO).

3b. US (ITAR / EAR) Export Control Assessment

☑ This document does not contain US origin Technical Data (Technology)

☐ This document contains "Technology" which is controlled by the U.S. government under *[USML category number / ECCN]* and which has been received by *[legal entity]* under the authority of *[licence number / ITAR exemption / EAR licence exception / NLR]*.

☐ This document contains technology which is designated as EAR99 (subject to EAR and not listed on the USML/CCL).

Note: Any re-export or re-transfer of this document in part or in whole must be made in accordance with the appropriate regulation (ITAR or EAR) and applicable authorization. If in any doubt please contact your local ECO.

3c. Technical Rater Information

This document has been assessed by the following Technical Rater:

Assessed and classified by: Sabine Pröll

Date classification completed: 04.03.2026

[1] "Controlled Technology" is defined as any Information necessary for the design, development, production, use, operation, maintenance or repair of export controlled goods. Examples of such Information are blueprints, plans, diagrams, models, engineering designs, manuals, requirements specifications and instructions etc. If in any doubt please contact your local Export Control Officer (ECO).

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

DOCUMENT CHANGE DETAILS

Issue	Date	Modified Element	Description of Change
01	04.03.2026	All	Initial version

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

Table of Contents

1 INTRODUCTION AND SCOPE 7
1.1 Introduction 7
1.2 Structure and Content of Document 7
1.3 Summary Description 7
1.3.1 MLI 8
1.3.2 MLI not part of this specification 9
1.3.3 FSSM & supports 9
2 DOCUMENTS 10
2.1 Applicable Documents 10
2.2 Reference Documents 10
2.3 Normative Documents 10
3 SPACECRAFT DESCRIPTION AND DEFINITIONS 11
3.1 Spacecraft Overall Description 11
3.2 Reference Frames 13
3.3 Acronyms 13
4 FUNCTIONAL AND PERFORMANCE REQUIREMENTS 15
4.1 General Functional Requirements 15
4.2 MLI performance 15
4.3 Flexible Second Surface Mirror (FSSM) Radiator Areas 17
5 DESIGN AND INTERFACE REQUIREMENTS 19
5.1 MLI Blanket Design 19
5.1.1 Accessibility 25
5.1.2 Transportation 26
5.2 Structural Interface Requirements 28
5.2.1 Mass 28
5.2.2 MLI Attachment 28
5.2.3 MLI Grounding Design 31
5.2.4 Mechanical Interface Control Documents 32
5.3 Ground Use & Safety Requirements 33
5.3.1 Special Handling Instructions 33
5.3.2 Interchangeability / Testability 33
5.3.3 Identification and Marking 33
5.4 Workmanship 34
5.5 Cleanliness and Contamination 34
5.6 Verification Requirements 35
5.7 Product Assurance Requirements 35
6 GENERAL DESIGN AND INTERFACE REQUIREMENTS 37
6.1 General Design and interface Requirements 37
6.1.1 General Design Requirements 37
6.1.1.1 Lifetime 37

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6.1.1.2 Design Safety 37
6.1.1.3 Dependability and Safety 38
6.1.1.4 Venting 38
6.1.1.5 Interchangeability 38
6.1.1.6 Identification & Marking 38
6.1.1.7 Accessibility/Maintainability 39
6.1.1.8 Transportation, Handling and Storage 39
6.1.1.9 Materials 40
6.1.1.10 Ground Support Equipments 41
6.1.2 Mechanical Design and Interface Requirements 41
6.1.2.1 Structural Design 41
6.1.2.2 Design Requirements 42
6.1.2.3 Mechanical Interface Control Documents 42
6.1.3 Thermal Design and Interface Requirements 43
6.1.3.1 Definition of Temperatures and Terms 43
6.1.3.2 Thermal Control 43
6.1.4 Electrical Design and Interface Requirements 44
6.1.4.1 EMC Requirements 44
6.2 Environment Design Requirements 45
6.2.1 Atmospheric Conditions 45
6.2.1.1 Humidity 45
6.2.1.2 Cleanliness 45
6.2.1.3 Pressure Environment 45
6.2.2 Mechanical Environment 45
6.2.2.1 Ground Operations Loads 45
6.2.3 Thermal Environment 46
6.2.3.1 On-ground phase 46
6.2.3.2 In-orbit phase 46
6.2.4 Radiation Environment 48
6.3 Unit Level Environment Test Requirements 48
6.3.1 General 48
6.3.1.1 Test Definition 48
6.3.1.2 Test Facilities Requirements 49

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

LIST OF FIGURES

Figure 3.1-1: S2NG Spacecraft - in deployed configuration 11
Figure 3.1-2: S2NG overall dimensions 12
Figure 3.2-1: S2NG Satellite Reference Frame 13
Figure 4.3-1: Cross Section of LIR with indicated SSM taping (TBC) 18
Figure 5.1-1: Areas requiring MLI coverage 21
Figure 5.1-2: MLI Type External 1 22
Figure 5.1-3: MLI Type External 2 23
Figure 5.1-4: MLI Type Internal 23
Figure 5.2-1: Nomenclature of Stand-Off Types 29
Figure 5.2-2: Blanket grounding with rivets 31
Figure 6.1-1: Temperature definition for unit thermal design (extract from ECSS-E-ST-31C) 43

LIST OF TABLES

Table 4.2-1: MLI performance 16
Table 4.3-1: Radiator Areas (TBC) 17
Table 5.1-1: Nominal MLI areas (TBC) 24
Table 5.1-2: Transport Environment Conditions (TBC) 27
Table 6.1-1: Mass Budget Maturity Margins 42
Table 6.2-1: Limit Accelerations for Ground Operations 45
Table 6.2-2: Unit Transportation Limit Shock Load 46
Table 6.2-3: Unit Non-Op, Op and Switch-on Temperatures 47
Table 6.2-4: Solar constant 47

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

1 INTRODUCTION AND SCOPE

1.1 Introduction

This specification establishes all Thermal Hardware related technical requirements. 3

1.2 Structure and Content of Document

This document includes all technical requirements to the specified item. Section 4 includes the functional, performance and specific requirements for the item. Section 5 concerns the requirements relevant to the system, i.e. interfaces and design requirements, as well as verification requirements which define details about tests and analyses. 8

Each requirement has a headline giving the verification method and in cases where it is useful a key-word. On the left hand margin of this headline each requirement has a unique number. This number is independent from the requirements location in the document and it never changes. So a requirement can be clearly identified by reference to this number. Underneath this headline the requirement wording (and figures) are given. In some cases the requirement is complemented by a comment or descriptive text. A comment is separated by horizontal lines and has italic letters. Descriptive text is given in roman font and is marked by dashed horizontal lines. 9

The verification methods are: 11

- T Test
- A Analysis
- R Review of Design
- I Inspection
- S Similarity

1.3 Summary Description

S2NG-THW-22//R

The S2NG Thermal Hardware (THW) shall comply to the requirements herein to provide the S2NG spacecraft with the required thermal hardware and functions. It shall consist of the following items assembled together by the THW contractor.

S2NG-THW-2578//R

The detailed geometrical design of all thermal hardware shall be based on the CAD data provided in AD-13 and in accordance with the MLI IRD (TBD).

S2NG-THW-4274//R

The Supplier shall be responsible for the provision of the Spacecraft MLI and FSSM including stand-offs and clip washers in accordance with the MLI IRD (TBD).

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-4275//R

The Supplier shall be responsible for the provision of any adhesives or tapes for attachment of studs & close-out of blankets.

S2NG-THW-4276//R

The Supplier shall be responsible for the provision of grounding leads.

S2NG-THW-4277//R

The Supplier shall be responsible for the provision of the necessary transport / storage containers for the MLI, FSSM and any required material.

1.3.1 MLI

27

- Baseplate
- Propulsion Plate
- Thruster Assembly (4x)
- MY Panel
- PYPZ Panel
- PYMZ Panel
- MZPY Panel
- MZMY Panel
- PZPY Panel
- PZMY Panel
- Top Floor
- Propellant Tank Cone (internal)
- Propellant Tank (internal)
- Pressurant Tanks (2x, internal)
- Battery & Harness (2x)
- CESS (6x)
- S-Band Antenna Bracket (2x)
- Ka-Band Antenna Pointing Mechanism (APM)
- GNSS Antenna Bracket (2x)
- Solar Array HDRM Brackets (4x)
- Umbilical Connector Brackets (2x)
- Skin Connector Bracket(s)
- Hoisting Brackets (6x)
- Venting Hole Cover(s)

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

- FVD Brackets (2x)
- All necessary supports, brackets, assembly kits, fasteners and grounding wires to assemble and integrate the MLI and Radiator surfaces to the S/C

1.3.2 MLI not part of this specification

2855

- AMSI
- STR-OH
- GFU
- Skirt (between platform and payload)
- LCT
- Solar Array

1.3.3 FSSM & supports

24

- Launcher Interface Ring
- MY Panel
- PYPZ Panel
- PYMZ Panel
- MZPY Panel
- MZMY Panel
- PZPY Panel
- PZMY Panel
- Batteries

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

2 DOCUMENTS

2.1 Applicable Documents

The following documents are applicable documents to this specification.

ID	Document Number	Title
To be deleted	ADS.E.1054	CAD data exchange requirement specification
AD-01	S2NG-SOW-ADSF-SYS-1002223258	Thermal Hardware SSOW
AD-42	ADS.E.1050	MICD Specification & CAD Data Exchange Requirements
AD-13	S2NG-210000A00C (stp File)	Thermal Hardware Input CAD model
AD-08	S3NGT-SP-ADSF-1000262	S3NGT Product Assurance Requirements for Suppliers
AD-00	SPS_MSC-LI-1000181994	Contractual Applicable Documents List

2.2 Reference Documents

The following documents amplify or clarify the contents of this document.

ID	Document Number	Title
RD-03	ECSS-E-HB-32-21A	Adhesive Bonding Handbook for advanced structural materials

2.3 Normative Documents

The following normative documents contain provisions which apply to this subsystem specification. The scope of this provision shall be qualified by the requirement in this document that invokes the normative document e.g. by quoting the applicable section number or paragraph in the normative document.

For normative documents with specific issue numbers or dates, subsequent amendments to, or revision of any of these publications, do not formally apply to this document. However, parties are encouraged to investigate the possibility of applying the more recent editions of the normative documents. For undated references, the latest edition of the publication referred to applies.

ID	Document Number	Title
ND-01	ECSS-E-ST-10-02C	Verification
ND-02	ECSS-E-ST-10-03C	Testing
ND-03	ECSS-E-ST-31C	Thermal control general requirements
ND-04	ECSS-E-ST-10-04C	Space environment

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

3 SPACECRAFT DESCRIPTION AND DEFINITIONS

3.1 Spacecraft Overall Description

The Sentinel-2 Next Generation (S2NG) system thermal design is mainly driven by its orbital parameters (altitude, inclination & LTAN), the S/C attitude (required by the payload) and the dissipating equipment and instrument units needing a stable thermal environment and a way to get rid of their waste heat in any mission phase respective S/C mode.

The main body of the S/C consists of a square shaped central structure reaching from the Launcher Interface Ring (LIR) all the way up to the Platform top floor.

The optical payload (AMSI) is mounted on the Platform top floor via kinematic mounts.

The Platform MY panel carries the deployable Solar Array via the SADM as well as the PCDU which is located on the panel inside. Embedded heat pipes distributing the heat dissipation laterally in ±Z direction of the PCDU radiator.

(see also Figure 3.1-1 & Figure 3.1-2)

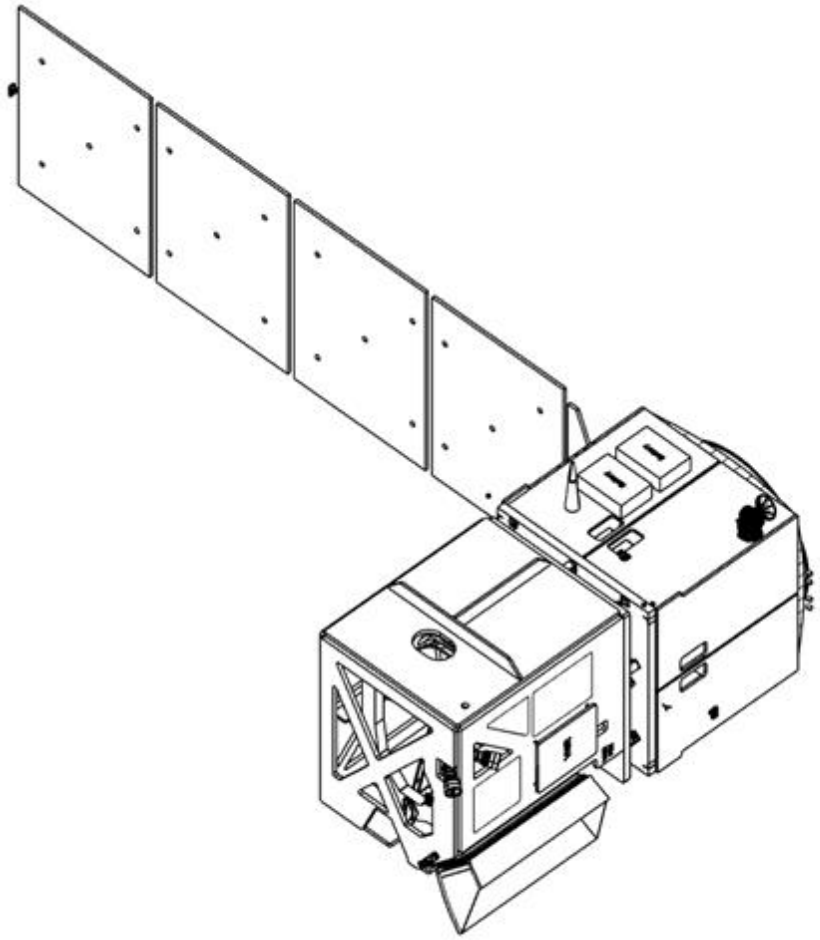


Figure 3.1-1: S2NG Spacecraft - in deployed configuration

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

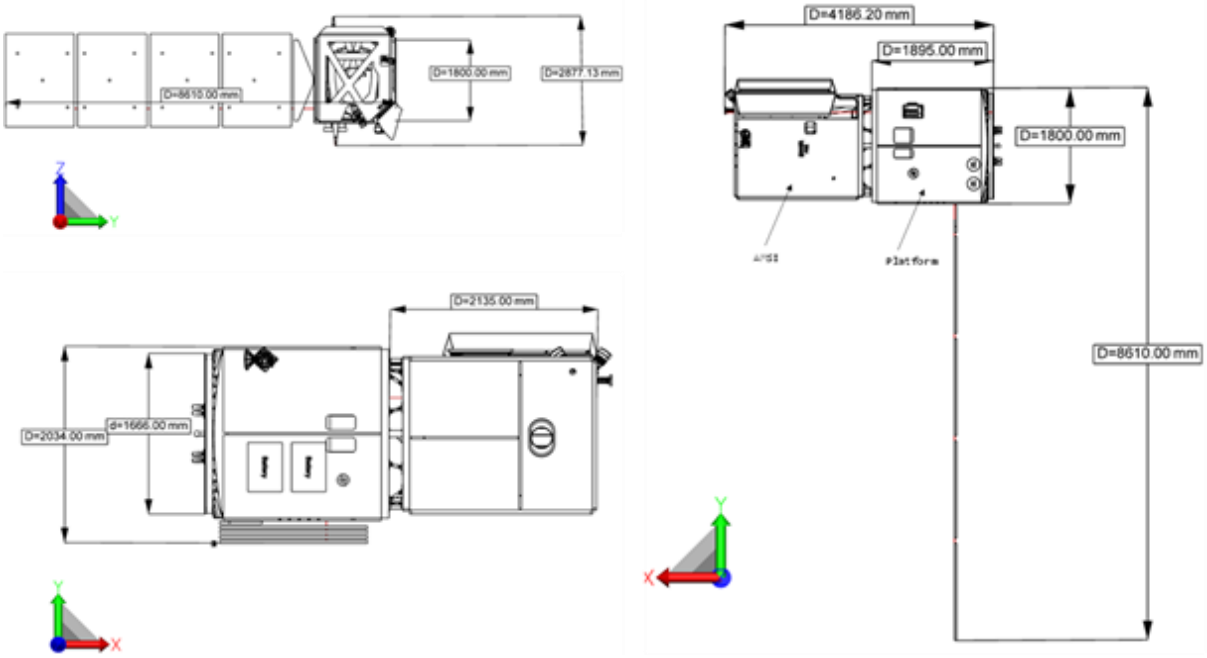


Figure 3.1-2: S2NG overall dimensions

As a result of the pointing (+Z to nadir and +X in flight direction) and orbit parameters, the sun illuminates during sunlit phase the spacecraft mainly from -Y (sun) and -Z (zenith) direction, but also hits the +X (front, payload location), -X (rear) and slightly +Z (nadir) sides. Only the +Y (deep space) side never faces the sun during nominal operations. Therefore, most of the thermally high dissipating units are located on the +Y deep space panels in order to provide the most efficient thermal environment for heat rejection. 942

The units heat rejection is mainly realized by a direct heat path through the supporting aluminium honeycomb panels and a SSM (Teflon) covered surface. 943

For an efficient spreading of the PCDU thermal dissipation across its radiator, embedded heat pipes are foreseen on the -Y (sun) panel with an orientation in Z-direction.

Units with a moderate to high thermal dissipation like PCDU, OBC, RIU, MMFU, etc. are located directly on S/C panels. Dedicated thermal doublers and SSM-covered radiator areas are foreseen to ensure sufficient heat spreading and heat radiation capability towards the external environment in order to fulfil the individual temperature requirements of the units. Thermally less sensitive units and/or units with lower heat dissipation like magnetorquers or reaction wheels are mainly accommodated on (secondary) internal structure panels. For these units the S/C internal surrounding environment is sufficient to keep them within their required thermal conditions for all mission phases. 1031

Two external battery modules are mounted thermally decoupled to the Platform +Z (nadir) panel, both equipped with an individual radiator pointing towards Earth. Each module shall be enclosed by a MLI. 1034

The propulsion module is a self-standing assembly which will be inserted through the Platform baseplate and thermally decoupled from the interior by thermal washers and MLI encapsulation. 1028

The AMSI mounted to the Platform top floor provides its own thermal control realized with MLI, radiators and internal heaters. 1022

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

3.2 Reference Frames

The S2NG Spacecraft reference frame is depicted in Figure 3.2-1:

- Origin: Center of the Launch interface ring in the separation plane between the launcher and the satellite
- +Xs: Longitudinal axis pointing from the origin towards S2NG satellite, collinear with the launcher velocity vector. Flight direction.
- +Zs: Oriented toward the Nadir direction
- +Ys: Transverse axis, completing the right handed orthogonal triad.

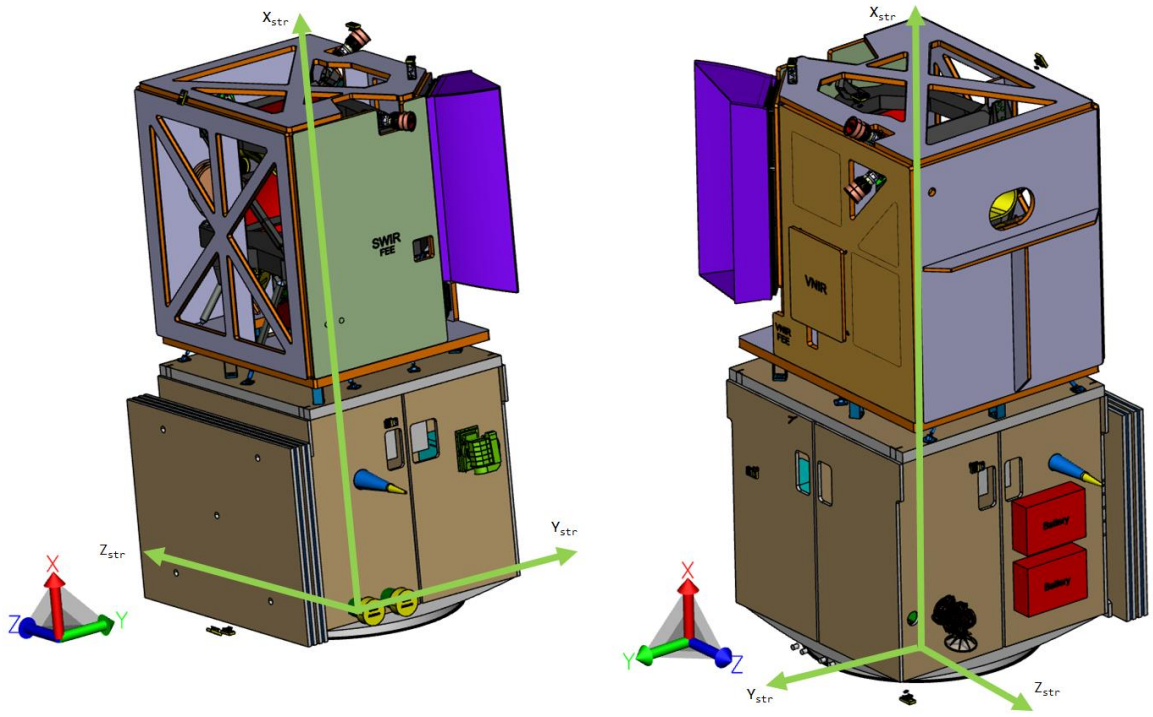


Figure 3.2-1: S2NG Satellite Reference Frame

3.3 Acronyms

The following table gives an overview about all used acronyms.

Abbreviation	Definition
AD	Applicable Document
AIT	Assembly, Integration and Testing
AIV	Assembly, Integration and Verification
AMSI	Advanced Multi-Spectral Imager
BOL	Beginning Of Life
CADL	Contractual Applicable Documents List
CDR	Critical Design Review
CESS	Coarse Earth and Sun Sensor
CFRP	Carbon Fibre Reinforced Polymer
DDV	Design, Development and Verification Plan
DSP	Deployable Solar Panel
ECSS	European Cooperation for Space Standardization
EOL	End Of Life
EQSR	Equipment Qualification Status Review

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

Abbreviation	Definition
ESA	European Space Agency
FM	Flight Model
FSSM	Flexible Second Surface Mirror
GDIR	General Design and Interface Requirements
GNSS	Global Navigation Satellite System
h	Heat transfer coefficient
HDRM	Hold Down and Release Mechanisms
ICD	Interface Control Document
ITO	Indium Tin Oxide
ITT	Invitation To Tender
LEOP	Launch and Early Orbit Phase
LIR	Launch Interface Ring
MAIT	Manufacturing, Assembly, Integration and Test
MLI	Multi-Layer Insulation
MRR	Manufacturing Release Review
NP	Nadir Panel
PLM	Payload Module
PROP	Propulsion System
RD	Reference Document
S/C	Spacecraft
SA	Solar Array
SLI	Single Layer Insulation
SOW	Statement Of Work
SSM	Second Surface Mirror
SVM	Service Module
TBC	To Be Confirmed
TBD	To Be Defined
Text	Temperature of external MLI layer
THR	Thruster
Tint	structure temperature
TR	Test Review
TRR	Test Readiness Review
UV	UltraViolet

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

4 FUNCTIONAL AND PERFORMANCE REQUIREMENTS

730

NOTE:

The thermal hardware items and their sizing requirements are all TBC, will be updated for kick-off and may vary during phase C/D until thermal hardware CDR.

4.1 General Functional Requirements

S2NG-THW-2252//R

The supplier shall propose and justify a general degradation factor for MLI blankets. This factor can be based on comparative tests from other projects.

S2NG-THW-4291//R

The design shall make use of flight proven materials and processes.

S2NG-THW-4292//R

The design shall promote the ease of analysis and verification.

S2NG-THW-4293//R

The design shall minimise uncertainties, e.g., sensitivity to design parameters.

S2NG-THW-4294//R

The design shall be optimised w.r.t. mass, reliability and cost.

4.2 MLI performance

S2NG-THW-109//T,S

The MLI including fixation parts shall withstand a general temperature range from -180°C to +180°C without deterioration of thermal or mechanical properties. The Thruster Assembly MLI shall withstand an extended temperature range of up to +340°C (intermittent, TBC).

S2NG-THW-2573//T,S

The design of the blanket fixation to structure shall withstand a temperature range of ±80°C at the mounting structure without deterioration of mechanical properties.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-828//R

All blankets adjacent to a radiator area shall provide an overall trimming capability of +/-20% of the radiating area.

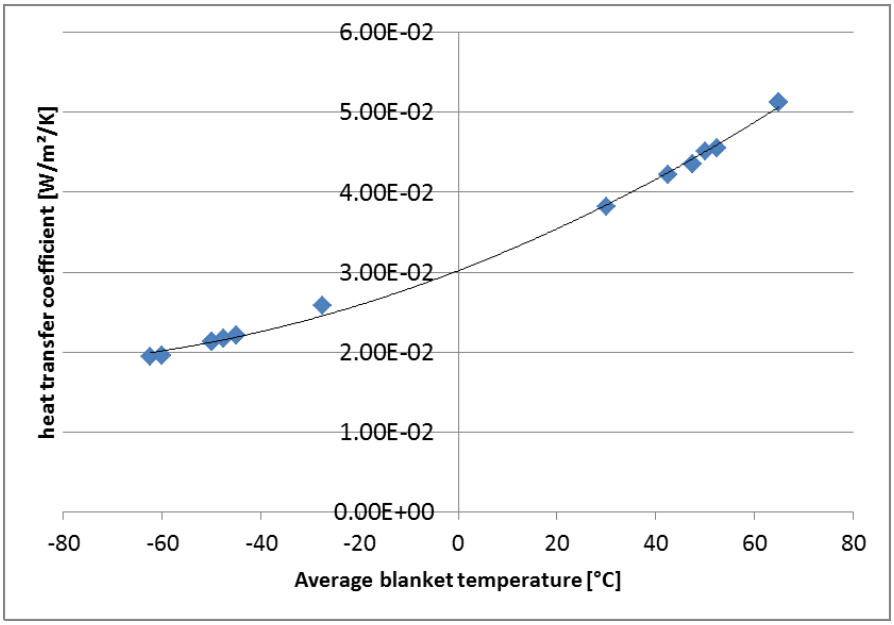
Note: Details of radiator trimming shall be agreed with the Customer before CDR.

S2NG-THW-111//T,S

The undisturbed MLI blankets shall have a heat transfer coefficient not greater than given in Table 4.2-1.

Table 4.2-1: MLI performance

T_avg (°C)	h [W/m²/K]
-62.5	1.95E-02
-60	1.96E-02
-50	2.13E-02
-47.5	2.17E-02
-45	2.21E-02
-27.5	2.58E-02
30	3.81E-02
42.5	4.22E-02
47.5	4.35E-02
50	4.51E-02
52.5	4.55E-02
65	5.12E-02



S2NG-THW-850//S

The hemispherical emissivity of the external layer of the external MLI shall be greater than 0.77.

S2NG-THW-851//S

The solar absorptivity of the external layer of the external MLI shall be less than 0.58 (EOL).

S2NG-THW-4295//S

The hemispherical emissivity of the external layer of the internal MLI shall be less or equal than 0.05.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

4.3 Flexible Second Surface Mirror (FSSM) Radiator Areas

S2NG-THW-2575//R

Radiator areas are required for S2NG to control the temperatures within applicable limits. SSM (ITO / 5 mil FEP / Ag / Inconel) tape shall be the baseline design of the radiators. The tape shall be grounded to the structure using an electrically conductive adhesive (e.g. 3M 9703).

The presently calculated radiator areas are given in Table 4.3-1. ***The quantities in this table are indicative and shall not be finally binding.***

Table 4.3-1: Radiator Areas (TBC) 3705

Radiator	Location	Dimensions (m)	Radiator Area (m²)
Ka-Band	PYPZ panel	0.60 x 0.75	0.45
MMFU	PYMZ panel	0.75 x 0.70	0.53
DPU	PYMZ panel	0.375 x 0.80	0.30
PCDU	MY panel	0.975 x 0.80	0.78
ICU A, ICU B	MY panel	0.23 x 0.70	0.16
OBC, RIU, APME	PZPY panel	0.925 x 0.70	0.65
STR-EUs	PZMY panel	0.25 x 0.47	0.12
SBT	MZMY panel	0.18 x 0.33	0.06
LCT-EU	MZPY panel	0.65 x 0.70	0.46
BAT1	Battery Module	0.508 x 0.354	0.18
BAT2	Battery Module	0.508 x 0.354	0.18
Total			**3.9 ± 50%**

S2NG-THW-894//R,S

The FSSM used for the radiator areas shall have a hemispherical emissivity of > 0.78 and a solar absorptivity of < 0.16 at EOL in orbit. BOL values shall be provided.

S2NG-THW-2253//R

To ensure electrical surface conductivity the SSM space-exposed surface shall be ITO coated.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-2574//T,S

The radiators including fixation parts shall withstand a temperature range from -45°C to +70°C without deterioration of thermal or mechanical properties.

In order to avoid potential entanglement of MLI and clamp band at upper stage release, the LIR shall be taped with SSM in vicinity of the launcher interface 2854

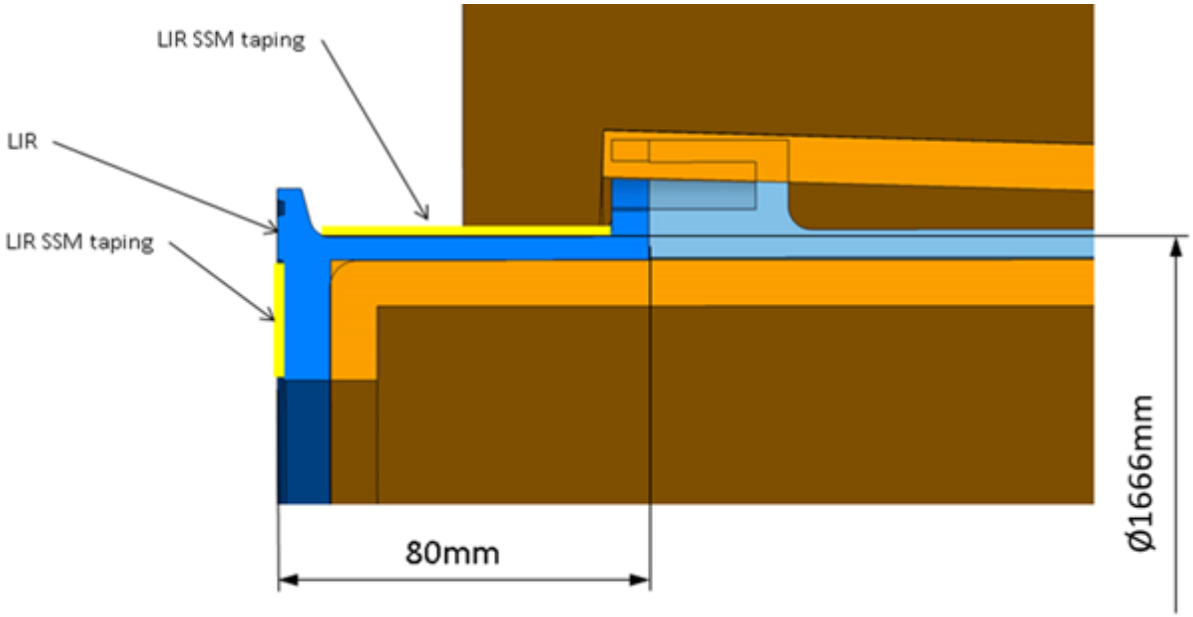


Figure 4.3-1: Cross Section of LIR with indicated SSM taping (TBC)

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

5 DESIGN AND INTERFACE REQUIREMENTS

S2NG-THW-3704//R

The overall detailed design of the passive thermal hardware (breakdown, sizing and shaping) shall be based on the CAD data delivered AD-13.

5.1 MLI Blanket Design

S2NG-THW-153//T,S

Unless test evidence from previous projects show that for the selected Mylar foil thickness shrinking is not expected, all Mylar foils thicker than 0.5 mil shall be pre-shrunk by keeping them for 4 hours at 110°C+/- 5°C in air at 1bar +/- 0.1bar.

S2NG-THW-2861//R

Areas where MLI has to enclose equipment only partly, like antennas, potential gaps between equipment and MLI shall be minimized. Adequate measures shall be foreseen, described (e.g. UM) for potentially needed, late adjustments.

S2NG-THW-4296//A,R

The MLI design shall provide adequate venting in installed configuration to preserve its integrity, preventing contamination to other parts.

S2NG-THW-4297//A

Venting provision shall be such that the MLI is able to withstand a pressure range of 1 bar to 10^{-8} bar.

S2NG-THW-147//I,R

The MLI layers shall be perforated with an adequate number of venting holes (holes/area) in order to allow the air to escape during depressurisation under the launcher fairing.

The perforation shall be selected such that contamination of S/C equipment by particles escaping through the perforation holes is avoided (offset of hole pattern of individual layers).

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-674//R

Venting of the S2NG compartments during launch and ascent shall be ensured in such a way that air is ejected towards ±Y panels through a TBD dimensioned venting aperture (final venting vector is TBD).

S2NG-THW-2298//A

The dimensioning of the venting zone/holes used for depressurization of the S/C cavity shall take into account an EMC tight mesh (INOX - stainless steel; thickness 50 µm; transparency of 65 %) reducing the effective cross section of the ventilation path.

S2NG-THW-3693//R

No constructions are allowed, which compress the blanket layers. Folds or bend constructions forming butt-joints are not allowed.

S2NG-THW-774//R

Potential heat leaks and/or sunlight trapping via venting devices shall be avoided or minimized.

S2NG-THW-4299//R

The design of the MLI shall be compatible with any orientation of the satellite to the sun.

S2NG-THW-3694//R

All slits and cut-outs shall be protected against tearing by providing the slit ends with a 3 mm diameter hole (crack stoppers) and shall be reinforced with a Kapton 2mil circular patch, sticked on to the external and internal layer, to avoid rupture of the foils.

S2NG-THW-704//R,S

All thermo-optical property requirements shall be applicable for End of Life (EOL).

S2NG-THW-706//R,S

The design of the MLI blankets shall be such that the blankets can be easily removed from and put back at least 6 times on their respective locations at the spacecraft without degradation of their material, mechanical, electrical and thermal characteristics during their life-time.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-4300//R

The Supplier shall asses and raise any exception on blankets number of installations, as required in S2NG-THW-706 at proposal.

S2NG-THW-4301//I,R

The external blankets shall be closed at the final integration with transfer tape on the internal side (preferred) or final taping on outer side between the stand-offs, to avoid sun intrusion except for those areas specifically considered and declared for venting.

The Platform external areas requiring MLI coverage are indicated in the following Figure 5.1-1 (highlighted in red frame). AMSI MLI is excluded.

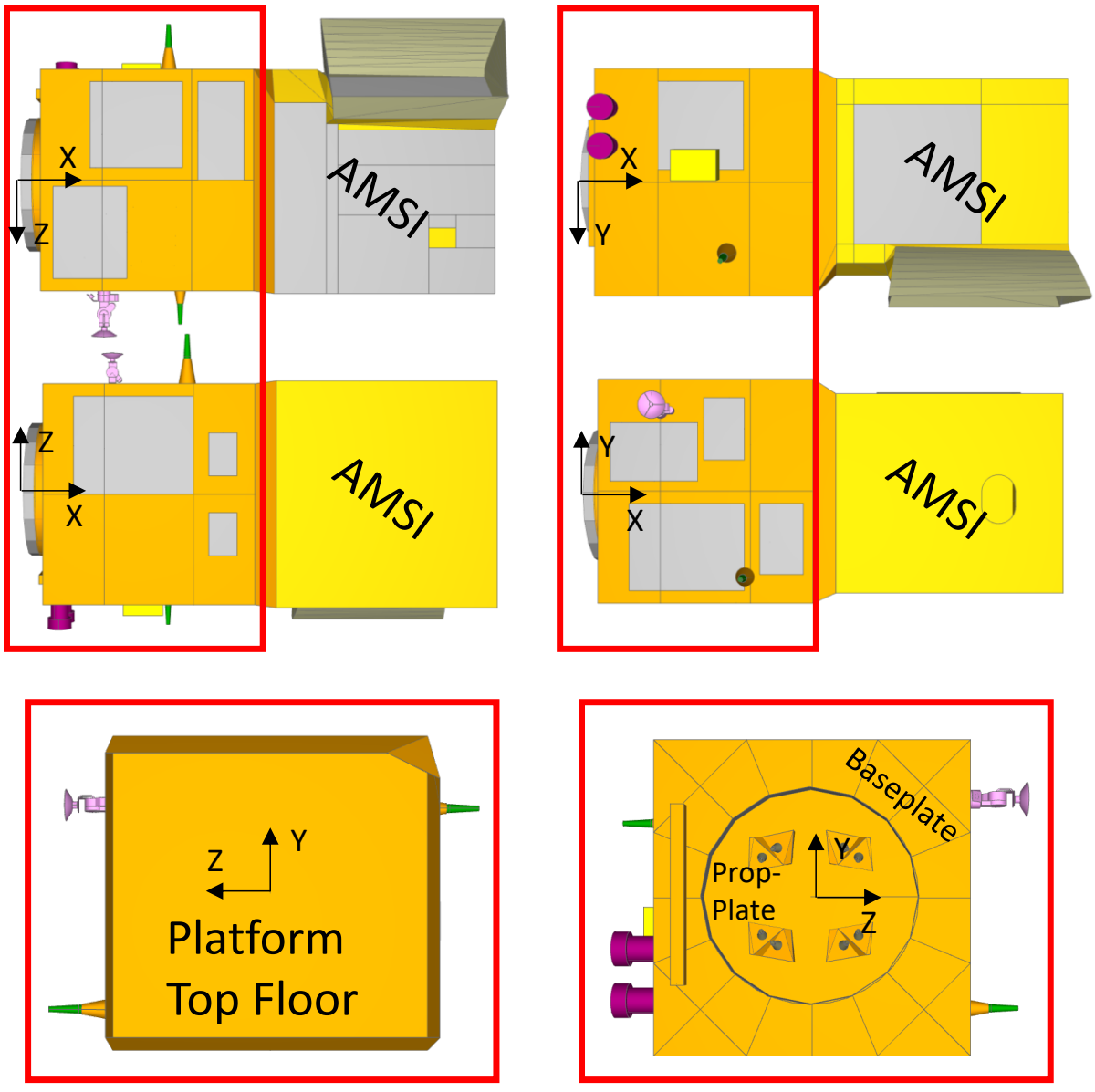


Figure 5.1-1: Areas requiring MLI coverage

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-3696//I,R

The standard external MLI exposed to space shall have the following composition (final composition to be proposed by supplier) Figure 5.1-2:

- *1 outer layer of 2 mil Kapton / VDA, perforated*
- *1 layer of VDA / 1 mil Kapton / VDA, perforated*
- *1 layer Dacron spacer;*
- *13 x*
  - *1 layer of VDA / 0.25 mil Mylar / VDA, perforated*
  - *1 layer of Dacron spacer*
- *1 inner layer of VDA / 1 mil Kapton / VDA, perforated*

3699

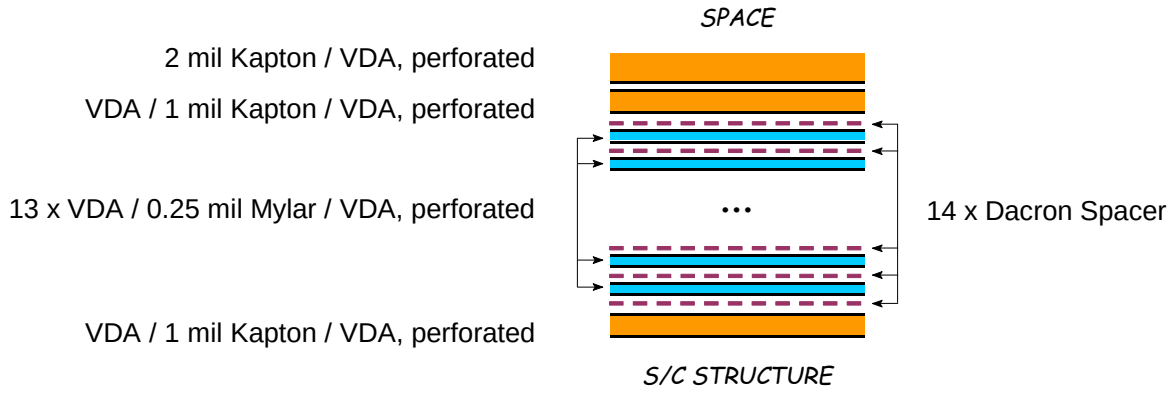


Figure 5.1-2: MLI Type External 1

S2NG-THW-3731//I,R

The external MLI for the thruster assembly shall have the following composition (final composition to be proposed by supplier) Figure 5.1-3:

- *1 outer layer of 2 mil Kapton / VDA, perforated*
- *1 layer Glassfibre Spacer GSW-25g;*
- *10 x*
  - *1 layer of VDA / 0.3 mil Kapton / VDA, perforated*
  - *1 layer of Glassfibre Spacer GSW-25g*
- *1 inner layer of VDA / 1 mil Kapton / VDA, perforated*

3732

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

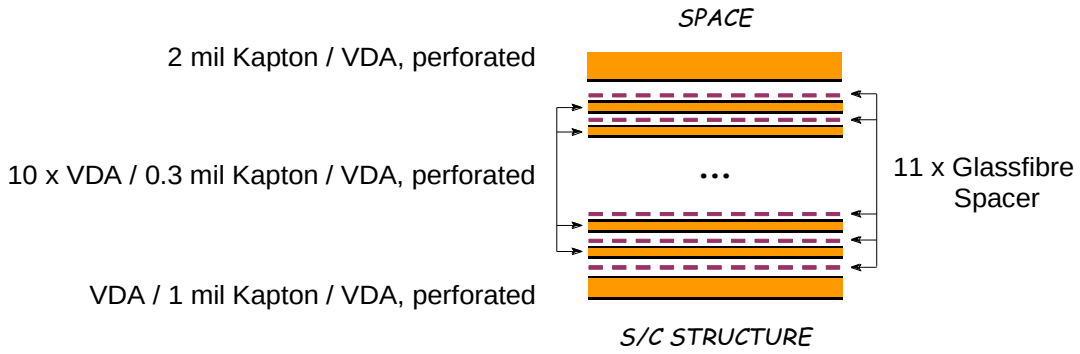


Figure 5.1-3: MLI Type External 2

S2NG-THW-3733//I,R

The internal MLI for Tank Cone, Propellant Tank, Pressurant Tank(s) and Top Floor shall have the following composition (final composition to be proposed by supplier) Figure 5.1-4:

- *1 layer of VDA / 1 mil Kapton / VDA, perforated*
- *1 layer Dacron spacer;*
- *13 x*
  - *1 layer of VDA / 0.25 mil Mylar / VDA, perforated*
  - *1 layer of Dacron spacer*
- *1 inner layer of VDA / 1 mil Kapton / VDA, perforated*

3734

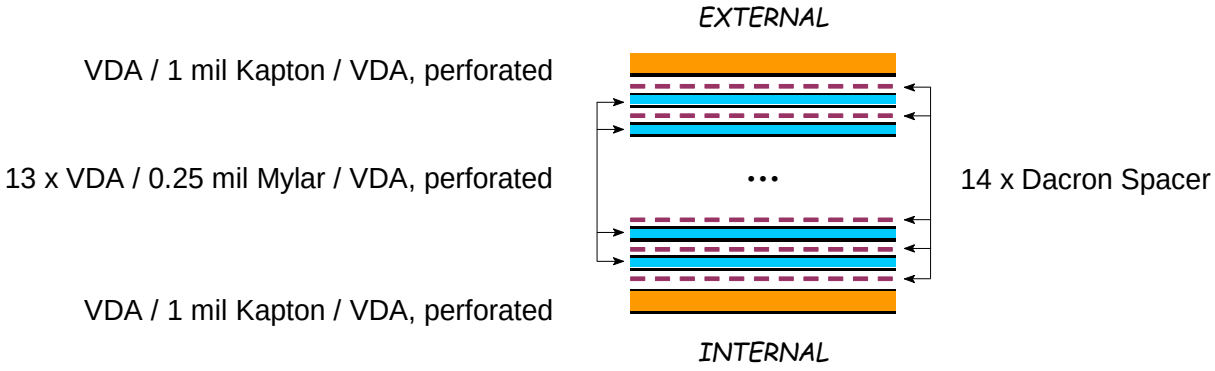


Figure 5.1-4: MLI Type Internal

S2NG-THW-3697//I,R

The blanket areas shown in Table 5.1-1 are the nominal areas (excluding overlaps). Adequate overlaps of minimum 50 mm to ensure a good thermal performance shall be provided.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

The quantities in this table are indicative and shall not be finally binding. *They do not take into account aspects such as limitations imposed by raw material size. The final number of the blankets shall be defined by the thermal hardware subcontractor and agreed with S2NG thermal control to consider AIT aspects.*

Table 5.1-1: Nominal MLI areas (TBC) 3698

Nr.	Location	Quantity	MLI Area per Item (m²)	MLI Type
1	Baseplate	1	1.0	EXT-1
2	Propulsion Plate	1	2.2	EXT-1
3	Thruster Assembly	4	0.1	EXT-2
4	MY Panel	1	2.2	EXT-1
5	PYPZ Panel	1	1.1	EXT-1
6	PYMZ Panel	1	0.75	EXT-1
7	MZPY Panel	1	1.1	EXT-1
8	MZMY Panel	1	1.5	EXT-1
9	PZPY Panel	1	0.9	EXT-1
10	PZMY Panel	1	1.4	EXT-1
11	Top Floor	1	3.2	INT
12	Propellant Tank Cone	1	1.5	INT
13	Propellant Tank	1	2.0	INT
14	Pressurant Tanks	2	0.2	INT
15	Battery & Harness	2	1.0	EXT-1
16	CESS	6	0.1	EXT-1
17	S-Band Antenna Bracket	2	0.1	EXT-1
18	Ka-Band Antenna Pointing Mechanism	1	0.4	EXT-1
19	GNSS Antenna Bracket	2	0.1	EXT-1
20	Solar Array HDRM Brackets	4	0.01	EXT-1
21	Umbilical Connector Brackets	2	0.05	EXT-1
22	Skin Connector Bracket(s)	1	0.08	EXT-1
23	Hoisting Brackets	6	0.1	EXT-1
24	Venting Hole Cover(s)	1	0.05	EXT-1
25	FVD Brackets	2	0.05	EXT-1
	Total		**24.0 ±20%**	

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

Additional Test MLI blankets (ca. 20 m²) are required for the TVAC test, e.g. for test adapter, around cable looms or for radiator trimming.

The MLI shall be designed to allow easy access to the following locations (e.g. by use of dedicated local blankets, MLI wrappings, flaps in the MLI or other suitable local design):

approx. 6 Hoisting points

approx. 7 Connector brackets

approx. 4 Solar Array hold-downs

5.1.1 Accessibility

S2NG-THW-757//R

Easy access to S/C equipment requiring maintenance shall be granted by the Thermal H/W up to last integration phases before mating to the launch vehicle.

S2NG-THW-759//R

For integration and test purposes, access to equipment located inside the spacecraft shall be granted by use of dismountable items.

S2NG-THW-761//R

Accessibility shall be provided to the connectors (or connector brackets) for harness mounting/dismounting by e.g. local blankets or easy fold back.

S2NG-THW-762//R

Accessibility shall be provided to the grounding hardware for mounting/dismounting.

S2NG-THW-763//R

Test connectors and optical cubes shall be easily reachable at integrated S/C levels.

S2NG-THW-4302//R

All alignment cube faces used for internal and end-to-end alignment, shall be visible without complete dismounting of flight hardware.

Note: Except for MLI on the cube itself during all phases of integration including the fully integrated satellite in launch configuration. Easy fold back of bigger blankets by opening a minimum number of fixations (clip washers) is also allowed.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-764//R

All AIT connectors will be mounted on a skin connector plate and shall be easily accessible when the S/C is fully integrated, in an unambiguous manner.

S2NG-THW-765//R

All safe and arm connectors shall be mounted on a skin connector plate and shall be easily accessible when the S/C is fully integrated, in an unambiguous manner.

S2NG-THW-4303//I,R

The external blankets shall be closed at the final integration with transfer tape on the internal side (preferred) or final taping on outer side between the stand-offs, to avoid sun intrusion except for those areas specifically considered and declared for venting.

5.1.2 Transportation

S2NG-THW-666//R

The specified items shall be shipped to the integration sites in reusable transport containers, which provide adequate protection against all environments to be expected during ground transport.

S2NG-THW-673//R

In the case of special handling or operating precautions being necessary these shall be included in the suppliers handling and transportation procedures.

S2NG-THW-4304//A,R

The MLI transport containers shall meet the following transport environmental conditions shown in Table 5.1-2.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

Table 5.1-2: Transport Environment Conditions (TBC) 4305

Outside Container on Ground	Outside Container in Cargo Compartment -Air transport	Inside Container The container design (carrying flight H/W) shall guarantee the following atmospheric conditions inside the container:
Temperatures:		
The external extreme conditions to be considered are the following:-40°C to +65°C during 12 hours	The temperatures can be within the range of 0°C up to 20°C during flight in the Cargo Compartment. The IATA regulations shall apply.	+0°C to +30°C
Extreme air temperature change (thermal shock) During a 1-hour period: Increase or decrease of air temperature of 10K		
Thermal radiation: Possible direct 12h Sun illumination with flux of 1000 W/m^2 simultaneously with a maximum external temperature not exceeding 40°C		
Humidity:		
0% to 100% with possible rain fall of 64 mm in 2 hours	The relative humidity in the Cargo Compartment is not controlled and depends on flight altitude and atmospheric conditions. At cruise altitude of 30,000ft the relative humidity drops down to approximately 10% RH. On the ground the humidity inside the aircraft compartment coincides with the external environmental humidity.	Less than 60% The dew point shall never be reached inside the container
Pressure:		
For transportation on road and sea, and for integration operations, the atmospheric pressure extremes for transportation between major locations of the program are: max. pressure = 1060 hPa (barometric pressure) min. pressure = 920 hPa For MGSE storage purposes, ambient pressure applies, unless an item specification calls for special conditions.	Maximum rates of change of pressure during air transportation (air freighting conditions):670 N/m^2/s (6,7 mbar/s). For emergency case, the decompression safety device (burst disk, burst flap) of the container shall be in accordance to the international IATA flight regulations	
Cleanliness:		
Uncontrolled. The air might be saline.	Uncontrolled	Better than 100.000

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

5.2 Structural Interface Requirements

5.2.1 Mass

S2NG-THW-3269//T

The total mass of the passive thermal hardware (MLI, SSM, fixation parts, thermal tapes, all electrical grounding hardware) shall be less than 12 kg (TBC) including margins.

S2NG-THW-4306//T

The MLI supplier shall determine the mass of the flight hardware by measurement.

5.2.2 MLI Attachment

S2NG-THW-2577//R

All bonding interfaces of MLI fixation shall be assumed as follows (TBC):

- *Aluminium treated with Surtec 650 for all panels, brackets, batteries*
- *Titanium (Ti6AI4V STA (3.7164.7)) for the propellant tank*
- *CFRP for the pressurant tanks*

Note1: The supplier shall demonstrate the qualification status at EQSR and finalize any additional needed qualification before CDR latest.

Note2: Removal of surface treatments is part of the attachment processes and shall be part of the qualification.

S2NG-THW-150//R

Unless otherwise specified the MLI shall be fixed with stand-off's to the mounting structure.

S2NG-THW-4307//R

The design shall avoid triple fixation points (stand-offs attaching more than two blankets at the same time).

S2NG-THW-4308//R

Whenever possible the design for all double fixation points shall foresee double stand-offs, which are glued to the structure.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-4309//R

In case of non-feasibility of glued double stand-offs for double fixation points the supplier must agree the alternative solution like double floating pin or single stand-off with the customer.

4310

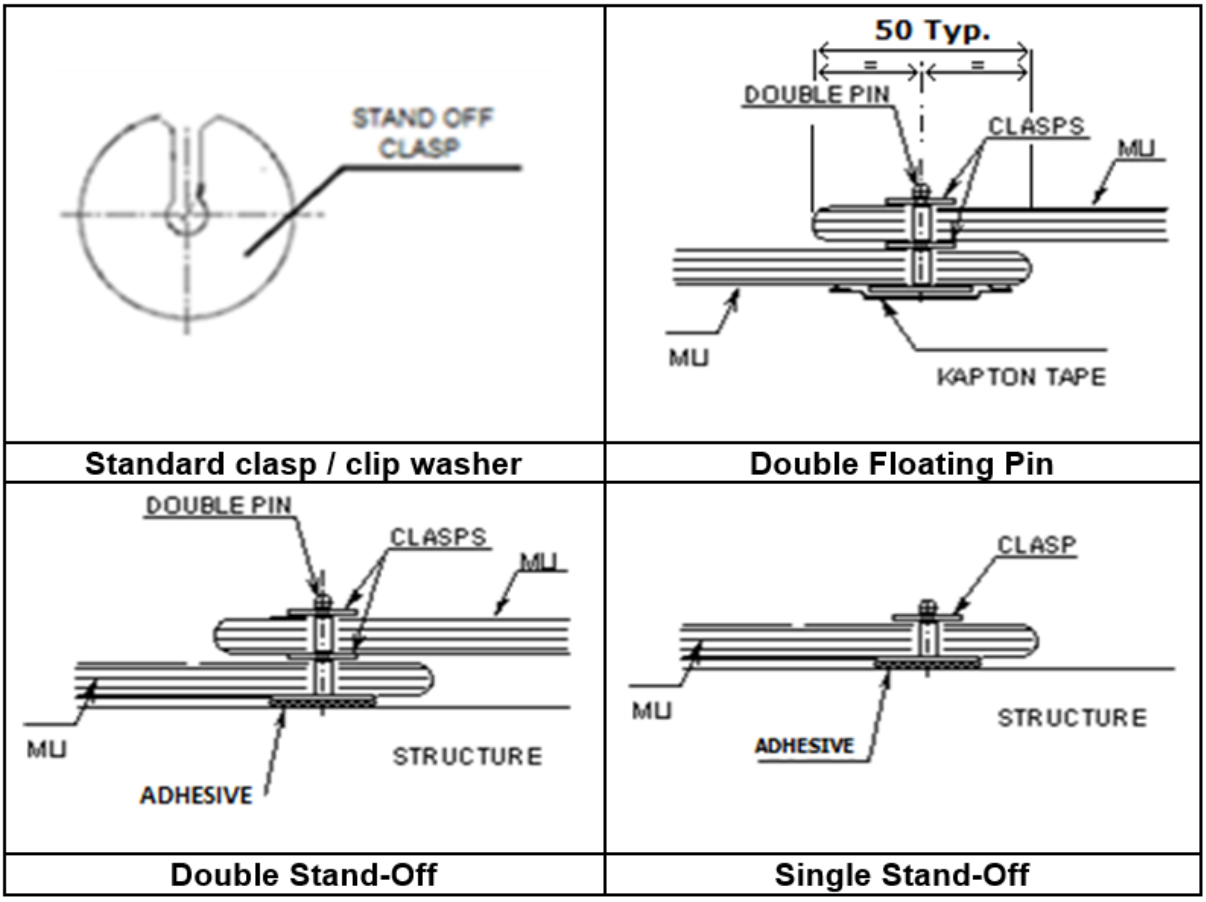


Figure 5.2-1: Nomenclature of Stand-Off Types

S2NG-THW-4311//R

The material for the stand-offs shall be aluminium, which is preferred, or Vespel SP1.

S2NG-THW-4312//R

Aluminium stand-offs shall be grounded to the structure beneath with conductive adhesive or by any other means, fully described and qualified.

Note: Alternative mounting options shall be agreed with the customer.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-4313//R

The dimensional requirement of stand-offs as well as the maximum intermediate distance between the stand-offs shall be determined by the supplier and agreed by Airbus.

S2NG-THW-151//R

For attachment of the MLI blankets to the supporting structure, a sufficient number of suitable stand-offs and clip washers shall be provided. The dimensional requirements of stand-offs (shaft diameter, base flange diameter, base flange thickness, maximum length and radius), as well as the maximum intermediate distance between the stand-offs shall be determined by MLI manufacturer and agreed by Airbus.

Any additional support structure needed to attach the blanket shall be provided by the subcontractor.

S2NG-THW-4314//R

Overlaps between individual MLI blankets shall be sufficient to ensure the performance of the overall MLI assembly on the satellite.

S2NG-THW-148//I,R

The overlapping sections of the MLI blankets shall be not less than 50 mm.

S2NG-THW-4315//R

The supplier shall agree any deviation from standard overlap with the customer.

S2NG-THW-748//R

The thermal hardware design shall allow simple mounting and dismounting procedures so that the instrument units can be installed or replaced late in the Spacecraft integration sequence.

S2NG-THW-3716//I,R

The blankets shall be attached to the panel without any gap between the blanket and the radiator.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-2859//R

MLI attachments in the vicinity of areas mentioned below (interfacing other MLI, not part of this specification) shall be designed to allow for the fixation of an additional, overlapping skirt MLI blanket.

- *all panels at the +X edge close to the top floor I/F for AMSI skirt MLI*

These areas will also be indicated in AD-13 and MLI IRD (TBD).

S2NG-THW-158//T,A,I

MLI blanket fixations consisting of stand-offs shall withstand a pull-off force of, at least, 100 N in stand-off - pin direction without any degradations. Sample tests shall be performed.

S2NG-THW-4316//T,A,I

MLI blanket fixations consisting on stand-offs shall withstand a pull-off force of, at least, 40 N in stand-off - shear direction without any degradations. Sample tests shall be performed.

S2NG-THW-4317//T

Minimum pull-off force in axial direction of a clip washer shall be 20 N.

5.2.3 MLI Grounding Design

S2NG-THW-4319//R

All blankets shall be grounded with minimum two fully independent grounding wires to the structure beneath, at grounding screwed points, as shown in Figure 5.2-2.

4320

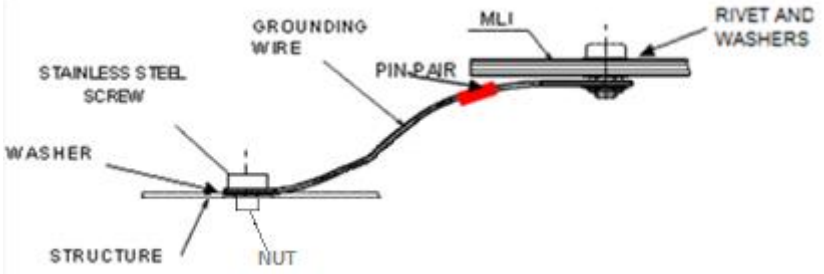


Figure 5.2-2: Blanket grounding with rivets

Every grounding point will give discharge path to several blankets. If more grounding points on the structure are required, it shall be communicated to the Customer

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-4321//R

The grounding wire shall consist of 2 parts connected by a pin socket connection.

S2NG-THW-4322//R

The pin socket connection location shall be close to the blanket, which allows an easy mounting and dismounting of the blanket.

S2NG-THW-4323//R

The grounding wire pin socket design shall foresee protection to prevent losing of the connection for flight.

S2NG-THW-4324//R

The grounding wires from MLI blanket to structure shall be such that the heat transferred by conduction is optimized for electrical and thermal performances.

Note1: electrical requirements are summarised in chapter 6.1.4

Note2: alternative grounding options need to be agreed with the customer.

5.2.4 Mechanical Interface Control Documents

S2NG-THW-4326//R

The MLI supplier shall detail and control the mechanical configuration including attachment points to the structure and between MLI with its interface requirements and dimensions in one or more Interface Control Drawing(s).

S2NG-THW-4327//R

Interfaces shall be subjected to a formal inspection, using interface data sheets in respect to mechanical properties. These data sheets shall be completed by the MLI Supplier and then included in the MLI Mechanical ICD.

S2NG-THW-4328//R

Interface Control Drawings shall be provided to the customer, with the following media and file formats (by order of preference):

- *CATIA V5.14*
- *STEP AP214*

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

- *2D DXF*

S2NG-THW-4329//R

The MLI Supplier shall agree the Interface Control Drawings with the customer.

Note: Any release or update of the ICD needs formal approval of the Customer in frame of a project review or by the Document Change Process.

5.3 Ground Use & Safety Requirements

S2NG-THW-779//R

The Thermal H/W shall be designed and fabricated with compatible materials in such a manner that all hazards associated with the spacecraft are eliminated, minimised and controlled.

5.3.1 Special Handling Instructions

S2NG-THW-781//R

In the case of special handling or operating precautions being necessary these shall be included in the suppliers handling and transportation procedures.

5.3.2 Interchangeability / Testability

S2NG-THW-783//R

Any Thermal H/W element shall be interchangeable without impact on Thermal H/W performances.

5.3.3 Identification and Marking

S2NG-THW-4331//I,R

The unit hardware shall be identified with a nameplate in order to achieve configuration traceability. The identification shall contain the following information:

- *Name of Manufacturer*
- *Project Name*
- *CI Number*
- *Serial Number*
- *Date of Manufacture*

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

Each Individual item shall be marked with a serial number. For standard parts and where the physical size of an item precludes identification of the hardware itself, a 'bag and label' technique shall be used (up to final integration) e.g. flying leads.

5.4 Workmanship

S2NG-THW-4333//R

All designs, manufacturing activities and equipment shall comply with the Product Assurance Requirements for Subcontractors AD-08 and the relevant space industry standards as listed in the CADL.

5.5 Cleanliness and Contamination

S2NG-THW-4336//

The cleanliness design requirements on the MLI blankets shall be as defined in the PA requirements AD-08.

S2NG-THW-4337//T,R

The MLI blankets shall be baked out and their cleanliness requirements monitored and maintained up to launch.

S2NG-THW-4338//R

The design of the MLI blankets shall avoid trapping of particles.

S2NG-THW-4339//I

The design of the MLI blankets shall ease cleaning.

Note: Following Bake-out, no solvent-based cleaning should be undertaken unless absolutely necessary.

S2NG-THW-4340//R

The design of the MLI blankets shall meet the MLI full performance under consideration of the cleanliness environment of the entire life-cycle up to EOL as given in S2NG-THW-4341.

S2NG-THW-4341//

The cleanliness environment for the MLI blankets of the entire life cycle consists of:

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

- *development, integration, and test phase at the supplier side*
- *handling and transport*
- *satellite AIT program and test phase*
- *launch preparation and launch*
- *in orbit commissioning and nominal operation*

S2NG-THW-4342//R

The MLI supplier shall provide a Cleanliness and Contamination Control Plan in accordance with ECSS-Q-70-01A (see CADL) considering the entire life-cycle up to EOL.

5.6 Verification Requirements

S2NG-THW-4344//R

The MLI supplier shall provide for each blanket a separate datasheet containing the following data as part of acceptance data pack (ADP).

- *Mass*
- *Material Composition*
- *Space Exposed Surface Absorptivity (α)*
- *Emissivity (ε)*
- *Specularity*

The thermo-optical properties are at the beginning of life (BOL).

5.7 Product Assurance Requirements

S2NG-THW-4346//I,R

All designs, manufacturing activities and equipment shall comply with the Product Assurance Requirements for Subcontractors AD-08 and the relevant space industry standards as listed in the CADL.

S2NG-THW-4347//R

All critical processes in view of thermal and mechanical interface requirements, the required mounting structure materials and surfaces treatments as well as environment and lifetime shall be clearly identified at the proposal for the project.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-4348//R

The MLI supplier shall take into account any qualification activities necessary in the development plan.

S2NG-THW-4349//R

The MLI supplier shall present all necessary qualification activities latest at EQSR.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6 GENERAL DESIGN AND INTERFACE REQUIREMENTS

4353

Applicability Log Information

Created by: Christopher.Speidel

Source Module Path: /Sentinel-3 NGT/Level 1_System/Sentinel NG GDIR Jacket Specification

Source Module Issue: 2.0

Source Filter: Appl-S2NG_MLI == Y

Source Filter Applied: true

Source Current View: Applicability - Platform equipments

Date: 02/26/26 12:52:53

6.1 General Design and interface Requirements

6.1.1 General Design Requirements

6.1.1.1 Lifetime

S2NG-THW-2601/GDI-135/A,R

The Unit shall be designed for on-ground lifetime of up to 10 years.

S2NG-THW-2602/GDI-137/A,R

The unit shall be designed to meet the requirements of its specification for an in-orbit lifetime of 7.5 years.

S2NG-THW-2603/GDI-138/R

Maintenance during storage shall be as limited as possible and, if required, shall be identified by the supplier for approval by the customer.

6.1.1.2 Design Safety

S2NG-THW-2605/GDI-141/A

General safety requirements for electronic units are as following:

- *All guards and covers provided for personal protection shall be clearly marked to indicate the voltage potential.*
- *Adequate shielding of control equipment and critical equipment is needed to prevent initiation of explosive devices from induced currents.*

S2NG-THW-2606/GDI-142/I

All units that could represent a hazard to personnel shall be clearly marked as such

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6.1.1.3 Dependability and Safety

6.1.1.4 Venting

S2NG-THW-2607/GDI-160/R

For all relevant thermal hardware, explicitly MLI, tapes and heatermats, venting provisions shall be incorporated.

S2NG-THW-2608/GDI-161/R

Unless a cavity is hermetically sealed adequate means of venting shall be provided in the design. The method of venting shall prevent the contamination of the cavity by the external environment and prevent the release of contaminants from the cavity.

6.1.1.5 Interchangeability

S2NG-THW-2611/GDI-164/R

All spacecraft units of the same part or configuration number shall be interchangeable in terms of form, fit and functions. The units must be of the same qualification status and reliability in order to meet the interchangeability requirement.

6.1.1.6 Identification & Marking

S2NG-THW-2613/GDI-168/I

For the particular case of connector identification, the following requirements shall apply:

- *Each unit or bracket is required to bear visible connector labels closely adjacent to the appropriate connector in order to allow a correct mating of the corresponding harness connector.*
- *For each unit or bracket, the connector identification shall be three alphanumeric characters:*
    a. *The first character is "J" for fixed (hard-mounted) connectors and "P" for mobile connectors.*
    b. *The two last characters consist of a 2 digits sequential number starting from 01.*
- *The location and content of the above described connector identification labels shall be included in the ICD of the relevant unit.*

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6.1.1.7 Accessibility/Maintainability

6.1.1.8 Transportation, Handling and Storage

6.1.1.8.1 Transport

S2NG-THW-2614/GDI-175/R

The units shall be transported using a container specifically designed to protect the flight hardware during ground or air transportation.

S2NG-THW-2615/GDI-176/R

The unit containers shall be instrumented and monitored against any damaging environment.

S2NG-THW-2616/GDI-177/R

The transport container shall ensure that the environment during transportation shall be maintained within the envelope of the defined launch mechanical environment (flight acceptance levels), i.e. transportation shall not drive the design.

S2NG-THW-2617/GDI-178/R

The storage container shall be designed to protect the unit without causing deterioration for the specified storage period. During long term storage the unit shall be stored under the following conditions:

- *Pressure: 970 mbar to 1050 mbar*
- *Temperature: 20°C ± 10°C*
- *Humidity: 45% ± 15%*
- *Cleanliness: Class 100,000 or better*

6.1.1.8.2 Unit Packing

S2NG-THW-2619/GDI-181/I

All units shall be packaged to ensure that it is sealed in a dry inert atmosphere using non-contaminating materials. The packing of the units shall be such that:

- *The pre-cleaned unit shall first be placed in a bag and sealed within,*
- *The protected unit shall then be placed in a second bag with a dehydrating agent and a label stating "OPEN IN A CONTAMINATION CONTROLLED ENVIRONMENT",*
- *The second bag shall also be sealed,*

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

- *The sealing of both bags shall be performed in cleanroom conditions (class 100,000 at least for the first bag) and a dry atmosphere.*
- *The double packaged units shall then be placed in a container that shall protect against all risk of degradation during transport and storage.*

6.1.1.8.3 Container Identification

S2NG-THW-2621/GDI-183/I

Each container shall be labelled, tagged or marked to show the following:

• Name Of Manufacturer

• Project Name

• Unit Name / Model

• Part Number

• Serial Number

• Airbus Purchase Order Number

• Quantity or weight (kg)

S2NG-THW-2622/GDI-184/I

In addition to the above, the container shall also be labelled with:

- *The statement "ONLY TO BE OPENED IN CLEANROOM CONDITIONS AFTER TEMPERATURE STABILISATION"*
- *Any recommendations necessary for the protection of the unit.*

6.1.1.9 Materials

6.1.1.9.1 Seals and Life limited items

S2NG-THW-2625/GDI-194/R

Any life limited items requiring periodic replacement during ground activities, and especially prior to launch, shall be identified to the Customer. The Unit supplier shall provide any procedures and special tooling required for replacement of seals.

6.1.1.9.2 Lubricants and Sealants

S2NG-THW-2627/GDI-196/R

No lubricants shall be used without the prior written agreement of the Customer.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6.1.1.9.3 Screw Locking

S2NG-THW-2629/GDI-198/I,R

All fasteners used on the unit shall be locked by adequate measures. This includes fixations of units onto the structure.

6.1.1.10 Ground Support Equipments

S2NG-THW-2631/GDI-201/I,R

Unit GSE shall be compatible with any model of a unit (EM, PFM, FM, Spare).

6.1.2 Mechanical Design and Interface Requirements

6.1.2.1 Structural Design

6.1.2.1.1 General Requirements

S2NG-THW-2635/GDI-208/A,R

The unit shall be designed to withstand the environments it will encounter during its lifetime without degradation of its performance, and without detrimental influence on the spacecraft or any other unit. The following shall be taken into account:

- *Fabrication and assembly loads (e.g. welding, interference fitting)*
- *Handling and transportation loads,*
- *Test loads (including thermal stresses),*
- *Launch loads (vibration (including shock), thermal and depressurisation),*
- *Operational loads (including thermal, attitude and orbit control induced loads).*
- *Structural dimensioning of the units shall consider critical combination of simultaneously acting loads (e.g. mechanical and thermal).*

6.1.2.1.2 Mass properties

S2NG-THW-2639/GDI-213/A

All mass estimations shall be accompanied by the definition of the design maturity of the concerned item (categories as per module or equipment SOW). The margin to be added to each item estimated mass for budget consolidation shall follow the rules expressed in Table 6.1-1.

The mass budget shall be reported to the customer through the datasheet as defined in APPENDIX A: MICD.

CATEGORY	DESIGN MATURITY	MARGINS
A/B	"Off-The-Shelf" weighted items	0%
A/B	"Off-The-Shelf" unweighted items	5%
C	"Off-The-Shelf" items requiring minor modifications	10%

2869

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

CATEGORY	DESIGN MATURITY	MARGINS
D	new designed / developed items, or items requiring major modifications or re-design	20%

Table 6.1-1: Mass Budget Maturity Margins

S2NG-THW-2641/GDI-240/T

The mass of an item must be measured with the following accuracy:

- *Item mass up to 10kg: +/- 0.010 kg*
- *Item mass from 10 to 20 kg: +/- 0,020 kg*
- *Item mass from 20 to 50 kg: +/- 0,040 kg*
- *Item mass from 50 to 100 kg: +/- 0,100 kg*
- *Item mass from 100 to 350 kg: +/- 0,200 kg*

S2NG-THW-2642/GDI-241/T

Any deviation between unit MCI measurement test and unit flight configuration shall be identified and reported. Resultant MCI figures for unit flight configuration shall be determined.

6.1.2.2 Design Requirements

6.1.2.3 Mechanical Interface Control Documents

S2NG-THW-2674/GDI-472/A

The mechanical configuration and its interface requirements and dimensions, shall be fully detailed in one (or more) Interface Control Drawing(s) that shall be fully referenced by the unit subcontractor.

This drawing shall detail all co-ordinate systems utilised and their relationship to each other, together with the principal unit interfaces.

The content of the mechanical ICD shall conform to "3D and 2D ICD content specification" (in accordance to the applicable CADL).

S2NG-THW-2675/GDI-473/I,R

Interfaces will be subjected to a formal inspection, using interface data sheets in respect to the "3D and 2D ICD content specification" (in accordance to the applicable CADL). These data sheets shall be completed by the unit subcontractor and then included in the EIDP.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-2676/GDI-480/R

Interface Control Drawings shall be provided as per "CAD data exchange requirement specification" (in accordance to the applicable CADL).

6.1.3 Thermal Design and Interface Requirements

6.1.3.1 Definition of Temperatures and Terms

3906

Qualification temperature range:

Temperature range at a TRP specified for the operating and non-operating mode and the switch-on condition of a unit, for which this unit is guaranteed to fulfil all specified requirements (see Figure 6.1-1)

3907

Acceptance temperature range:

Temperature range at a TRP obtained from the qualification temperature range after subtraction of qualification margins specified for the operating and non-operating mode and the switch-on condition of the unit (see Figure 6.1-1)

Note 1: the acceptance temperature range is the extreme temperature range that the unit can be allowed to reach, but never exceed, during all envisaged mission phases.

Note 2: the acceptance temperature range is the temperature range used during acceptance tests to verify all specified requirements and workmanship.

3908

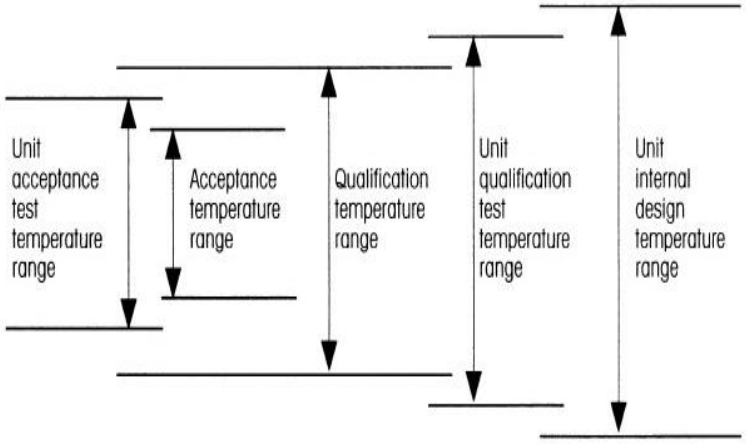


Figure 6.1-1: Temperature definition for unit thermal design (extract from ECSS-E-ST-31C)

6.1.3.2 Thermal Control

6.1.3.2.1 General

S2NG-THW-2682/GDI-577/T,A

The equipment thermal control must be testable on ground.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6.1.4 Electrical Design and Interface Requirements

6.1.4.1 EMC Requirements

6.1.4.1.1 Bonding

6.1.4.1.1.1 Mechanical Parts

S2NG-THW-2684/GDI-2187/T,R

Mechanical parts without electrical nor shielding function shall show a bonding resistance of less than 100 kOhm between any adjacent parts and the local GRR / metallic structure.

S2NG-THW-2685/GDI-2188/T,R

Mechanical Parts used for electromagnetic shielding shall show a bonding resistance of less than 2.5 mOhm measured under 1 Amp. between any adjacent parts and the local GRR (or connector bracket) / metallic structure.

6.1.4.1.2 Insulating materials

S2NG-THW-2687/GDI-2200/R

Space exposed insulating materials having a surface resistivity higher than 1E+9 Ω/square or having a bulk resistivity higher than 1E+13 Ω.m shall not be used.

6.1.4.1.3 Thermal parts

6.1.4.1.3.1 MLI

S2NG-THW-2690/GDI-2203/I

MLI shall carry at least 1 conductive layer and each conductive layer shall be bonded to structure to avoid electrical charge differential.

S2NG-THW-2691/GDI-2204/I,R

The grounding points shall be in a minimum of 2 locations (opposite corners). Each blanket shall be individually grounded to the structure. For small MLI pieces below 0.01m^2, bonding at one point is sufficient.

S2NG-THW-2692/GDI-2205/T

The resistance between each bonding point and the structure shall be lower than 100mΩ.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-2693/GDI-2206/T

The resistance between each bonding point and each point of a conductive layer shall be lower than 100Ω.

6.2 Environment Design Requirements

6.2.1 Atmospheric Conditions

6.2.1.1 Humidity

S2NG-THW-2697/GDI-2455/R

The unit shall be designed to withstand a relative humidity of 65% maximum during testing and transport.

6.2.1.2 Cleanliness

6.2.1.3 Pressure Environment

S2NG-THW-2700/GDI-2462/A,R

Units mounted on the S/C shall be designed to withstand without degradation, a de-pressurization rate of 70mbar/s maximum and a Delta-P of 150mbar over ambient.

6.2.2 Mechanical Environment

6.2.2.1 Ground Operations Loads

S2NG-THW-2703/GDI-2470/A,R

The units and associated transport containers shall be sized to survive transportation and handling loads as defined in Table 6.2-1.

Vertical and horizontal loads shall be considered as acting simultaneously (un-attenuated input to MGSE).

GROUND EVENT	VERTICAL (*)	HORIZONTAL
Ground Transportation	± 3.0g	± 2.0g
Air Transport	**± 5.0g**	**± 2.0g**

(*) Vertical is the direction parallel to the gravity

Table 6.2-1: Limit Accelerations for Ground Operations

S2NG-THW-2719/GDI-2481/A,R

During ground transportation the units shall withstand without damage the limit shock loads defined in Table 6.2-2.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

Table 6.2-2: Unit Transportation Limit Shock Load 4354

Direction	Amplitude (g)	Half Period (msec)	Pulse Shape	Number of Pulses
±X	±4.0	20	Saw Tooth	1
±Y	±4.0	20	Saw Tooth	1
±Z	±4.0	20	Saw Tooth	1

2720

S2NG-THW-2754/GDI-2516/T,A

All MGSE, containers, transportation and handling methods shall be such that flight hardware for which it is intended, never experience any environmental condition outside the defined envelope above

S2NG-THW-2756/GDI-2518/T,A

For containers, the shock requirement is a drop of 100 mm onto concrete of one corner of the container, with another corner of the container lying on the concrete floor.

The transportation containers shall be designed to ensure that the flight unit contained within is protected from and shall be undamaged by shocks.

Frequency	100 Hz	2kHz	10kHz
Shock response spectrum (Q=10) at unit interface	20g	2000g	2000g

4028

6.2.3 Thermal Environment

6.2.3.1 On-ground phase

S2NG-THW-2779/GDI-2987/T,A

For AIV and storage, the unit shall withstand temperature conditions as follows:

- *Short term Storage(prior or after transportation : -40°C to +60°C*
- *Integration : +10°C to +30°C*
- *Unit testing : Unit non-operational and operational temperature requirement ranges as defined in the unit specification, or by default as per Table 6.2-3.*

6.2.3.2 In-orbit phase

6.2.3.2.1 Temperature requirements

S2NG-THW-2783/GDI-2991/T,A

The unit shall be designed to withstand without degradation the qualification temperatures and acceptance temperatures defined in the unit specification, or by default in Table 6.2-3.

°C	Min Op	Max Op	Min Non-Op	Max Non-Op	Min Start up
Acceptance Temperature	-25	+55	-35	+65	-30

2816

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

°C	Min Op	Max Op	Min Non-Op	Max Non-Op	Min Start up
Qualification Temperature	-30	+60	-40	+70	-30

Table 6.2-3: Unit Non-Op, Op and Switch-on Temperatures 2838

6.2.3.2.2 Thermal radiation

S2NG-THW-2785/GDI-4775/A,R

When accommodated externally and submitted directly to thermal flux radiation, unit shall be designed to withstand the thermal environment encountered in its mission, as described hereafter.

6.2.3.2.2.1 Solar constant

S2NG-THW-2787/GDI-4778/A

The thermal control of any external unit accommodated outside the satellite, shall take into account the Solar flux as defined in Table 6.2-4.

2788

DATE	SOLAR IRRADIANCE (W/m²)
Winter solstice	1423
Summer solstice	1321
Equinox (autumnal)	1357

Table 6.2-4: Solar constant 2789

6.2.3.2.2.2 Earth albedo

S2NG-THW-2791/GDI-4799/A

The thermal control of any external unit accommodated outside the satellite, shall take into account the Earth albedo. The solar radiation reflected from the Earth is obtained with an average albedo value of 0.30 ± 0.05

6.2.3.2.2.3 Earth emitted radiation

S2NG-THW-2793/GDI-4800/A

The thermal control of any external unit accommodated outside the satellite, shall take into account the Earth emitted radiation. The value of infra-red Earth emission is 200W/m2 min (Earth temperature = 244 K) and 275 W/m2 max (Earth temperature = 264 K).

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6.2.3.2.3 Vacuum

6.2.3.2.4 Atomic Oxygen Environment

S2NG-THW-2797/GDI-5543/T,A

Any external unit shall withstand erosion and oxidation induced by monoatomic oxygen of a fluence

of 1.01E+21 at/cm^2 over the mission lifetime defined in SNG-S2NG-THW-2602.

Coating thickness (for example MLI) shall present adequate margins.

6.2.4 Radiation Environment

S2NG-THW-4043/GDI-6221/T,A,R

Each unit shall comply with the space radiation environment defined in [Sentinel-3 NGT: Space Environment Specification, rf. CADL, and the related radiation hardness requirements stipulated by the ECSS-Q Standard [ECSS-Q-ST-60-15C], scaled to the specified mission lifetime according to SNG_S2NG-THW-2602.

6.3 Unit Level Environment Test Requirements

6.3.1 General

6.3.1.1 Test Definition

6.3.1.1.1 Qualification tests

S2NG-THW-2805/GDI-3330/R

Qualification of the design shall be accomplished using representative flight configuration hardware and software.

The objective of the qualification shall be to demonstrate the capability of all hardware items to provide all specified performances under all environment and interface conditions and to survive their full operational life.

The test specimen shall be subjected to qualification loads. The test conditions shall not exceed design margins of safety nor excite unrealistic failure modes.

Items which can be qualified by an applicable qualification history, by similarity to qualified items, or purely by analysis, may require only limited qualification testing or even none. However, all safety critical items must be fully qualified by test.

Qualification testing shall be conducted at the relevant level according to the requirements defined in this section.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6.3.1.1.2 Acceptance Tests

S2NG-THW-2806/GDI-3332/T,A,R

Environmental acceptance testing shall be performed on all deliverable flight and flight spare hardware.

The objectives of acceptance testing are:

- *Demonstrate freedom from manufacturing and workmanship errors.*
- *Demonstrate that hardware and software performance comply with design specifications.*

During environmental acceptance testing, conditions or effects, similar to the mission environment shall be simulated.

Acceptance testing shall be conducted at the relevant level according to the requirements defined in this section.

6.3.1.2 Test Facilities Requirements

S2NG-THW-2808/GDI-3336/R

Any test facility to be used within the instrument assembly or unit test programme shall be capable to perform the required test within the specified limits and shall not impact the test objective or degrade the test article performance.

6.3.1.2.1 Accuracy of Test Instrumentation

S2NG-THW-2810/GDI-3338/R

The accuracy of instrument and test equipment used to control or measure the test parameters shall be in general, one order of magnitude better than the tolerance for the variable to be measured

Exceptions shall be specified in the relevant specifications and shall be agreed by the customer.

All instrumentation to be used for qualification and acceptance tests shall be subjected to approved calibration procedures and shall be within the normal calibration periods at the time of test. Instrumentation which will run out of calibration during the planned test time shall be not used.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

6.3.1.2.2 Tolerances of Test Parameters

S2NG-THW-2812/GDI-3340/R

The allowed test condition tolerances shall be applied to the specified nominal test values. Unless otherwise specified, the maximum allowable tolerances on test conditions or measurements shall be as per the tables in the following subsections. The tolerance on test parameters specifies the maximum allowable range within which the specified test level (input level) or measurement (output) may vary and excludes instrument accuracy.

6.3.1.2.2.1 Temperature

Temperature Range	Minimum Temperature	Maximum Temperature
-200°C to -50°C	-4°C / 0°C	0°C / +4°C
-50°C to +100°C	-3°C / 0°C	0°C / +3°C
+100°C to +370°C	-4°C / 0°C	0°C / +4°C

2898

6.3.1.2.2.2 Pressure

Pressure Range	Pressure
Above $1.3x10^{2}$ Pa (1 Torr)	±15%
$1.3x10^{-1}$ to $1.3x10^{2}$ Pa	±30%
Less than $1.3x10^{-1}$ Pa	±80%

2916

6.3.1.2.2.3 Relative Humidity

Relative Humidity	±5%

2930

6.3.1.2.2.4 Acceleration

Acceleration	0 / +10%

2935

6.3.1.2.2.5 Vibration Frequency

Random	±5% or 1Hz (whichever is greater)
Sine	±2% from 10 to 2000Hz

2940

6.3.1.2.2.6 Vibration Level

Sine Vibration:	Amplitude	±10% g peak
	Sweep Rate	±5%
Random Vibration Acceleration:		
Power Spectral Density	**Qualification**	**Acceptance**
20 to 500Hz (25Hz or narrower)	-1.0dB / +3.0dB	-3.0dB / +1.5dB
500 to 2000Hz (50 Hz or narrower)	-1.0dB / +3.0dB	-3.0dB / +1.5dB
Random Overall g RMS	±10%	±10%

2948

6.3.1.2.2.7 Acoustic Noise

Range	Sound Pressure Level
1/3 octave band (centre frequency)	-1.0dB / + 3.0dB
Overall	±1.5dB

2977

6.3.1.2.2.8 Test Duration Time

Test Duration Time	0 / + 10%

2988

6.3.1.2.2.9 Shock

2993

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

Response spectrum amplitude (1/6 octave centre frequency)	- 3 dB / + 6 dB 50 % of the SRS amplitude above 0 dB
Shock duration:	
<= 20 ms	0 / +20%
> 20 ms	0 / +10%
Shock Level	0 / +20%

6.3.1.2.2.10 Solar simulation

3010

Solar intensity distribution in reference plane	±4%	
Solar intensity distribution in reference volume	±6%	
Solar intensity stability	±1%	
Solar intensity stability (absolute)	±3%	
Type	**Wavelength (Angstrom)**	**Percent of Total Energy**
Far Ultraviolet	1 to 2000	0.008
Near Ultraviolet	2000 to 3800	6.995
Visible	3800 to 7000	39.88±TBD%
Near Infrared	7000 to 10000	22.59
Infrared	10000 to 20000	24.02
Far Infrared	20000 to 100000	6.45

6.3.1.2.2.11 Balancing

3052

Static	<0.25mm
Dynamic	<±0.1°

6.3.1.2.2.12 Others

3060

Force (static)	-0 / +3%
Centre of Gravity	See Section 6.1.2.1
Moment of Inertia	See Section 6.1.2.1
Forces and Moments (dynamic)	-0 / +3%
Pressure	±5% of max. specified value
Flow Rate	±5%
Leakage	$\pm 10^{-4}$ scc/s He at 1013hPa pressure differential if not otherwise specified
Mass	See Section 6.1.2.1.2
Dimensions (length)	< 0.125mm
Angular Measurements	0.5 arc min. with respect to each axis of the reference system of the test facility if not otherwise specified

6.3.1.2.3 Facility Ambient Condition

S2NG-THW-4045/GDI-3541/R

Unless otherwise specified herein, all measurements and tests shall be made at room ambient atmospheric pressure, temperature and relative humidity conditions, whereby 22 ± 3°C and 55 % ± 10 % RH shall not be exceeded. Whenever these conditions must be closely controlled in order to obtain reproducible results a reference temperature of 21°C, a relative humidity of 50 % and an atmospheric pressure of 1013 hPa shall be used together with whatever tolerances are required to obtain the desired precision of measurement. Actual ambient test conditions shall be recorded periodically during the test period.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

The cleanliness requirements for the test items during assembly, integration, transport and test shall be in accordance with the PA requirements.

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

End of Document

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

Requirement/Section Cross Reference

Page Numbers are the pages where the sections start

S2NG-THW-22 1.3 7
S2NG-THW-109 ..... 4.2 15
S2NG-THW-111 ..... 4.2 16
S2NG-THW-147 ..... 5.1 19
S2NG-THW-148 ..... 5.2.2 30
S2NG-THW-150 ..... 5.2.2 28
S2NG-THW-151 ..... 5.2.2 30
S2NG-THW-153 ..... 5.1 19
S2NG-THW-158 ..... 5.2.2 31
S2NG-THW-666 ..... 5.1.2 26
S2NG-THW-673 ..... 5.1.2 26
S2NG-THW-674 ..... 5.1 20
S2NG-THW-704 ..... 5.1 20
S2NG-THW-706 ..... 5.1 20
S2NG-THW-748 ..... 5.2.2 30
S2NG-THW-757 ..... 5.1.1 25
S2NG-THW-759 ..... 5.1.1 25
S2NG-THW-761 ..... 5.1.1 25
S2NG-THW-762 ..... 5.1.1 25
S2NG-THW-763 ..... 5.1.1 25
S2NG-THW-764 ..... 5.1.1 26
S2NG-THW-765 ..... 5.1.1 26
S2NG-THW-774 ..... 5.1 20
S2NG-THW-779 ..... 5.3 33
S2NG-THW-781 ..... 5.3.1 33
S2NG-THW-783 ..... 5.3.2 33
S2NG-THW-828 ..... 4.2 16
S2NG-THW-850 ..... 4.2 16
S2NG-THW-851 ..... 4.2 16
S2NG-THW-894 ..... 4.3 17
S2NG-THW-2252 ..... 4.1 15
S2NG-THW-2253 ..... 4.3 17
S2NG-THW-2298 ..... 5.1 20
S2NG-THW-2573 ..... 4.2 15
S2NG-THW-2574 ..... 4.3 18
S2NG-THW-2575 ..... 4.3 17
S2NG-THW-2577 ..... 5.2.2 28
S2NG-THW-2578 ..... 1.3 7
S2NG-THW-2601 ..... 6.1.1.1 37
S2NG-THW-2602 ..... 6.1.1.1 37
S2NG-THW-2603 ..... 6.1.1.1 37
S2NG-THW-2605 ..... 6.1.1.2 37
S2NG-THW-2606 ..... 6.1.1.2 37
S2NG-THW-2607 ..... 6.1.1.4 38
S2NG-THW-2608 ..... 6.1.1.4 38
S2NG-THW-2611 ..... 6.1.1.5 38
S2NG-THW-2613 ..... 6.1.1.6 38
S2NG-THW-2614 ..... 6.1.1.8.1 39
S2NG-THW-2615 ..... 6.1.1.8.1 39
S2NG-THW-2616 ..... 6.1.1.8.1 39
S2NG-THW-2617 ..... 6.1.1.8.1 39
S2NG-THW-2619 ..... 6.1.1.8.2 39
S2NG-THW-2621 ..... 6.1.1.8.3 40
S2NG-THW-2622 ..... 6.1.1.8.3 40
S2NG-THW-2625 ..... 6.1.1.9.1 40
S2NG-THW-2627 ..... 6.1.1.9.2 40
S2NG-THW-2629 ..... 6.1.1.9.3 41
S2NG-THW-2631 ..... 6.1.1.10 41
S2NG-THW-2635 ..... 6.1.2.1.1 41
S2NG-THW-2639 ..... 6.1.2.1.2 41
S2NG-THW-2641 ..... 6.1.2.1.2 42
S2NG-THW-2642 ..... 6.1.2.1.2 42
S2NG-THW-2674 ..... 6.1.2.3 42
S2NG-THW-2675 ..... 6.1.2.3 42
S2NG-THW-2676 ..... 6.1.2.3 43
S2NG-THW-2682 ..... 6.1.3.2.1 43
S2NG-THW-2684 ..... 6.1.4.1.1.1 44
S2NG-THW-2685 ..... 6.1.4.1.1.1 44
S2NG-THW-2687 ..... 6.1.4.1.2 44
S2NG-THW-2690 ..... 6.1.4.1.3.1 44
S2NG-THW-2691 ..... 6.1.4.1.3.1 44
S2NG-THW-2692 ..... 6.1.4.1.3.1 44
S2NG-THW-2693 ..... 6.1.4.1.3.1 45
S2NG-THW-2697 ..... 6.2.1.1 45
S2NG-THW-2700 ..... 6.2.1.3 45
S2NG-THW-2703 ..... 6.2.2.1 45
S2NG-THW-2719 ..... 6.2.2.1 45
S2NG-THW-2754 ..... 6.2.2.1 46
S2NG-THW-2756 ..... 6.2.2.1 46
S2NG-THW-2779 ..... 6.2.3.1 46
S2NG-THW-2783 ..... 6.2.3.2.1 46
S2NG-THW-2785 ..... 6.2.3.2.2 47
S2NG-THW-2787 ..... 6.2.3.2.2.1 47
S2NG-THW-2791 ..... 6.2.3.2.2.2 47

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.

S2NG-THW-2793 ..... 6.2.3.2.2.3 47
S2NG-THW-2797 ..... 6.2.3.2.4 48
S2NG-THW-2805 ..... 6.3.1.1.1 48
S2NG-THW-2806 ..... 6.3.1.1.2 49
S2NG-THW-2808 ..... 6.3.1.2 49
S2NG-THW-2810 ..... 6.3.1.2.1 49
S2NG-THW-2812 ..... 6.3.1.2.2 50
S2NG-THW-2859 ..... 5.2.2 31
S2NG-THW-2861 ..... 5.1 19
S2NG-THW-3269 ..... 5.2.1 28
S2NG-THW-3693 ..... 5.1 20
S2NG-THW-3694 ..... 5.1 20
S2NG-THW-3696 ..... 5.1 22
S2NG-THW-3697 ..... 5.1 23
S2NG-THW-3704 ..... 5 19
S2NG-THW-3716 ..... 5.2.2 30
S2NG-THW-3731 ..... 5.1 22
S2NG-THW-3733 ..... 5.1 23
S2NG-THW-4043 ..... 6.2.4 48
S2NG-THW-4045 ..... 6.3.1.2.3 51
S2NG-THW-4274 ..... 1.3 7
S2NG-THW-4275 ..... 1.3 8
S2NG-THW-4276 ..... 1.3 8
S2NG-THW-4277 ..... 1.3 8
S2NG-THW-4291 ..... 4.1 15
S2NG-THW-4292 ..... 4.1 15
S2NG-THW-4293 ..... 4.1 15
S2NG-THW-4294 ..... 4.1 15
S2NG-THW-4295 ..... 4.2 16
S2NG-THW-4296 ..... 5.1 19
S2NG-THW-4297 ..... 5.1 19
S2NG-THW-4299 ..... 5.1 20
S2NG-THW-4300 ..... 5.1 21
S2NG-THW-4301 ..... 5.1 21
S2NG-THW-4302 ..... 5.1.1 25
S2NG-THW-4303 ..... 5.1.1 26
S2NG-THW-4304 ..... 5.1.2 26
S2NG-THW-4306 ..... 5.2.1 28
S2NG-THW-4307 ..... 5.2.2 28
S2NG-THW-4308 ..... 5.2.2 28
S2NG-THW-4309 ..... 5.2.2 29
S2NG-THW-4311 ..... 5.2.2 29
S2NG-THW-4312 ..... 5.2.2 29
S2NG-THW-4313 ..... 5.2.2 30
S2NG-THW-4314 ..... 5.2.2 30
S2NG-THW-4315 ..... 5.2.2 30
S2NG-THW-4316 ..... 5.2.2 31
S2NG-THW-4317 ..... 5.2.2 31
S2NG-THW-4319 ..... 5.2.3 31
S2NG-THW-4321 ..... 5.2.3 32
S2NG-THW-4322 ..... 5.2.3 32
S2NG-THW-4323 ..... 5.2.3 32
S2NG-THW-4324 ..... 5.2.3 32
S2NG-THW-4326 ..... 5.2.4 32
S2NG-THW-4327 ..... 5.2.4 32
S2NG-THW-4328 ..... 5.2.4 32
S2NG-THW-4329 ..... 5.2.4 33
S2NG-THW-4331 ..... 5.3.3 33
S2NG-THW-4333 ..... 5.4 34
S2NG-THW-4336 ..... 5.5 34
S2NG-THW-4337 ..... 5.5 34
S2NG-THW-4338 ..... 5.5 34
S2NG-THW-4339 ..... 5.5 34
S2NG-THW-4340 ..... 5.5 34
S2NG-THW-4341 ..... 5.5 34
S2NG-THW-4342 ..... 5.5 35
S2NG-THW-4344 ..... 5.6 35
S2NG-THW-4346 ..... 5.7 35
S2NG-THW-4347 ..... 5.7 35
S2NG-THW-4348 ..... 5.7 36
S2NG-THW-4349 ..... 5.7 36

Airbus Defence and Space GmbH owns the copyright of this document which is supplied in confidence and which shall not be used for any purpose other than that for which it is supplied and shall not in whole or in part be reproduced, copied, or communicated to any person without written permission from the owner.